

FINAL YEAR PROJECT • 2025 — 2026

ForsaDrive

A Tunisian-context carpooling platform — Web, Mobile, and a unified backend

Presented by

Youssef BEN ABID & Anas YOUNES

Supervised by

Mr. Khalil SELMI (ATOMIC IT)

Ms. Ines BEN NASR (Academic)

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What we delivered and what comes next

Mobility in Tunisia — a real, daily problem

Public transport is irregular. Private cars travel half empty. Existing apps are not built for the local market.

Public transport

Overcrowded buses, irregular schedules, weak coverage of secondary cities.

Informal carpooling

Facebook groups, no identity verification, no payment trace, no rating.

International apps

BlaBlaCar/inDrive/Bolt — payment models and language not adapted to Tunisia.

PROBLEM STATEMENT

Tunisia lacks a secure, affordable digital ride-sharing solution adapted to local payment habits, available on web and mobile, and capable of building genuine trust between drivers and passengers.

ATOMIC IT — host company

- ▶ IT engineering services: web and mobile development, software consulting.
- ▶ Modern stack: Linux/Windows, .NET, Java, Android/iOS, PHP.
- ▶ Located in Kelibia, Nabeul. Director: Mr. Khalil SELMI.
- ▶ Flat structure — short feedback loops between developers and management.
- ▶ Internship inside the engineering team, mentored by senior developers.
- ▶ Agile Scrum applied: daily stand-ups, weekly reviews with the supervisor.

AT A GLANCE

Activity	Web & mobile development
Director	Mr. Khalil SELMI
Location	Kelibia, Nabeul
Phone	+216 55 343 224
Email	contact@atomicitpro.com
Methodology	Agile / Scrum

What people use today

Criterion	Facebook	inDrive	Bolt	BlaBlaCar	ForsaDrive
Identity verification	—	Partial	✓	✓	✓
In-app payment	—	—	✓	✓	✓
Long-distance rides	✓	—	—	✓	✓
Rating system	—	✓	✓	✓	✓
Adapted to local context	✓	Partial	Partial	—	✓
Student discount (50%)	—	—	—	—	✓
Wallet-based balance	—	—	—	—	✓

No existing option combines local fit with structured trust mechanisms. ForsaDrive sits exactly in that gap.

ForsaDrive — what we built

An integrated ecosystem: web app, mobile app, and a shared backend, covering the full lifecycle from registration to post-trip rating.

Local fit

50% prepayment + cash settlement,
French/English/Arabic with RTL, Tunisian-
recognized university domains.

Trust

Identity & student verification, ratings on both
sides, complaint workflow, reliability score.

Smart features

Compatibility score, driver analytics, ride
boosting, social feed, in-app chat, AI-assisted
helpdesk.

3

coordinated apps

4

functional sprints

20+

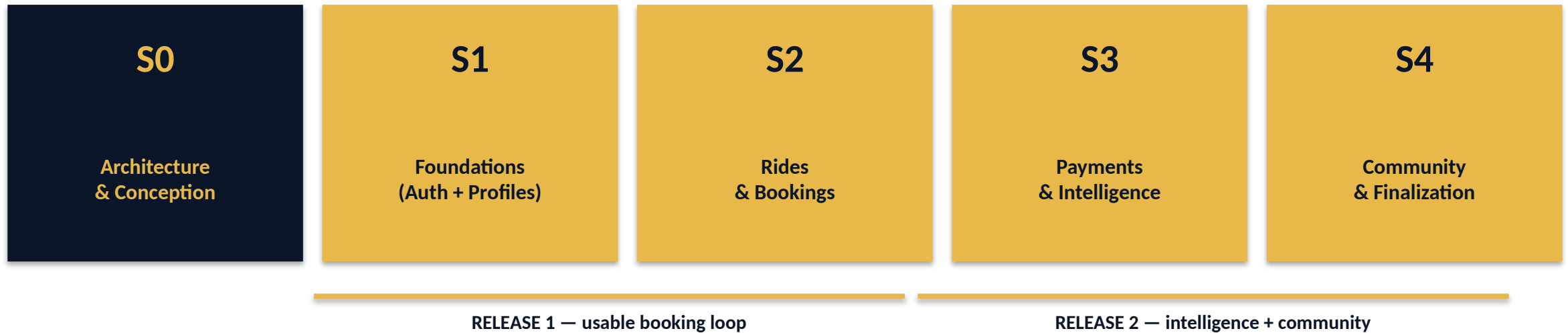
domain entities

3

supported languages

Scrum — five iterations grouped into two releases

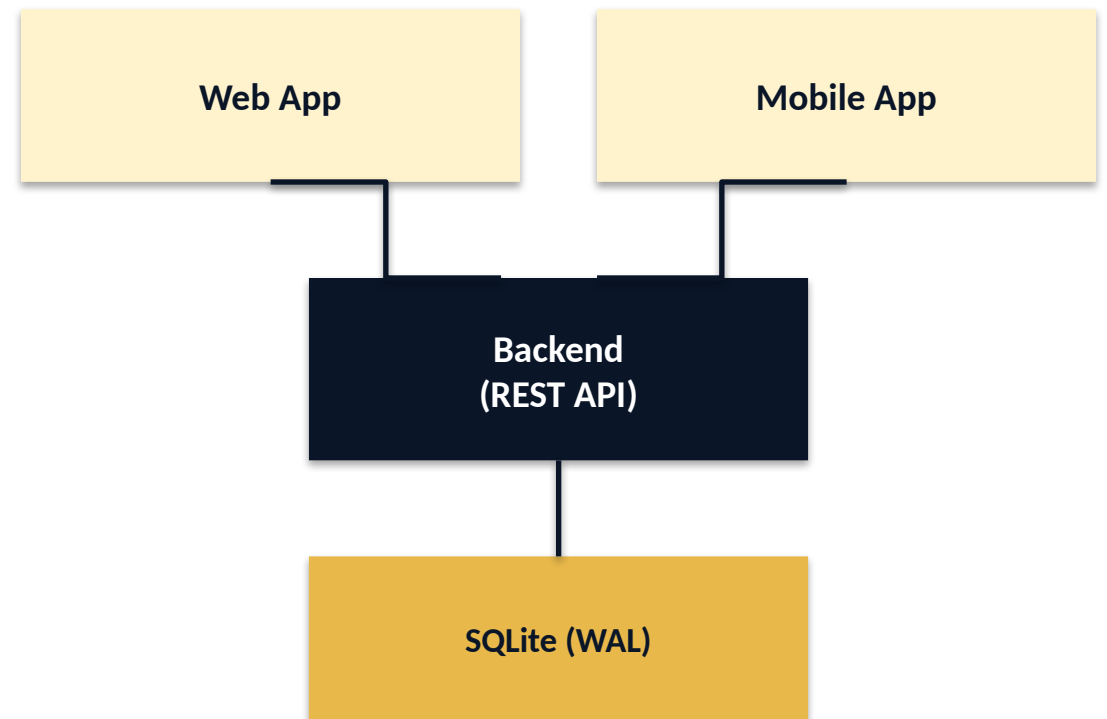
Sprint 0 prepared the ground. Four functional sprints then delivered two coherent releases of the product.



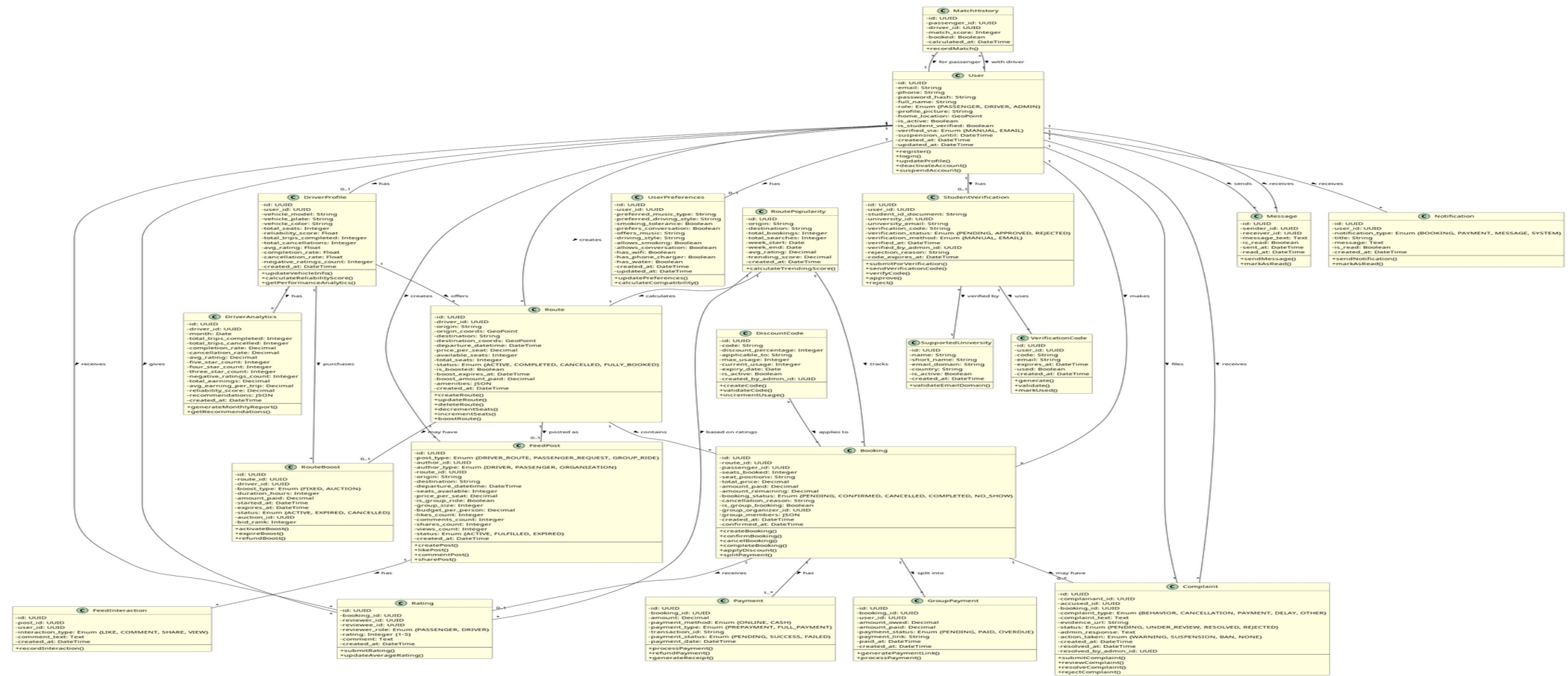
- ▶ Daily stand-ups inside the team — weekly reviews with the supervising engineer.
- ▶ Each sprint produced a usable increment, reviewed and demonstrated.
- ▶ Trello board kept the backlog visible — story points on a Fibonacci scale.

Architecture — client / server, three tiers

- ▶ Two clients (web + mobile) consume one REST API.
- ▶ Backend in PHP, exposed under /api/, secured with bearer tokens.
- ▶ Three tiers: presentation, application, data.
- ▶ External services (payment, notifications, file storage) only reached through the backend — no client secrets.
- ▶ HTTPS in production. Passwords with bcrypt. Audit log on sensitive operations.

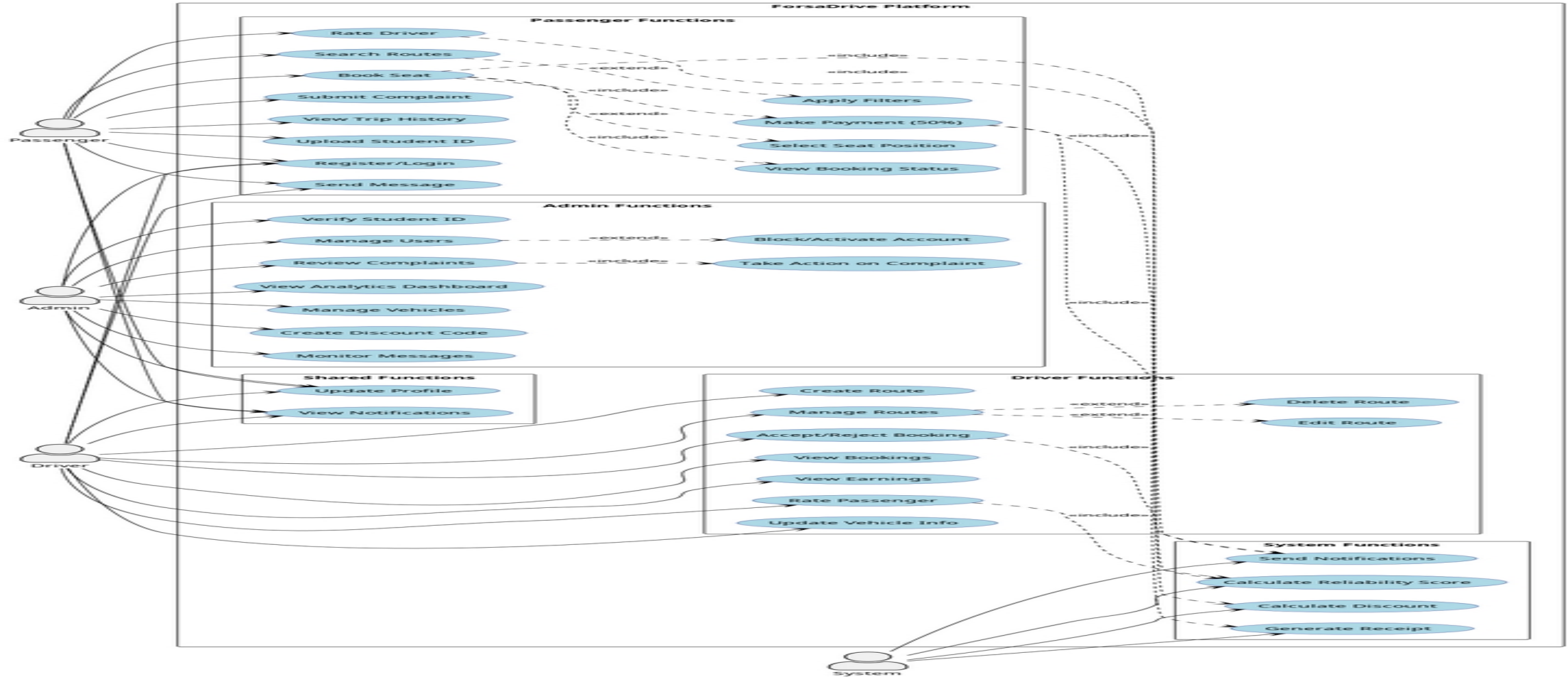


Class diagram – 20+ entities, 5 logical groups



Users & Profiles • Trips & Bookings • Payments & Ratings • Verification & Discounts • Communication & Moderation

Global use case — four actors



Passenger • Driver • Administrator • System (automated)

SPRINT 1

Foundations

MISSION

Account creation, authentication, profile, student verification, driver application & admin approval.

Sprint backlog (extract)

#	User story	Pts
US1.1	Register an account	5
US1.2	Login with email + password	3
US1.3	View / update profile	3
US1.4	Student verification via university email OTP	8
US1.5	Apply to become a driver (license + vehicle)	8
US1.6	Admin reviews driver applications	5
US1.7	Admin manages recognized university domains	3

Deliverables

- ▶ Self-service OTP flow (10-min code, 5-attempt cap, 30s cooldown).
- ▶ Driver application creates DriverProfile + Vehicle on approval.
- ▶ Audit log entry on every verification event.

SPRINT 2

Rides & Bookings

MISSION

Driver publishes a ride. Passenger searches, filters, books with 50% prepayment.

Sprint backlog (extract)

#	User story	Pts
US2.1	Publish a ride (origin, destination, date, price, seats)	5
US2.2	Modify or cancel a ride	3
US2.3	Search rides by origin / destination / date	5
US2.4	Filter by price, departure time, driver rating	5
US2.5	View ride detail and driver profile	3
US2.6	Book seats with 50% prepayment	8
US2.7	Group booking for friends	5
US2.8	Driver accepts / rejects booking requests	5

Deliverables

- ▶ Booking persisted only if prepayment succeeds — no orphan bookings.
- ▶ Search response indexed on origin, destination, departure_date.
- ▶ Activity diagram covered cash collection at end of trip.

SPRINT 3

Payments & Intelligence

MISSION

Wallet, promo codes, ride boosting, driver analytics, compatibility score.

Sprint backlog (extract)

#	User story	Pts
US3.1	View wallet balance and history	5
US3.2	Top up wallet	3
US3.3	Auto-apply 50% student discount on booking	3
US3.4	Apply organizational promo code	3
US3.5	Boost a ride to top of search/feed	5
US3.6	Driver dashboard with reliability + revenue + ratings	8
US3.7	Compatibility score badge per ride	5
US3.8	Recompute reliability after every completed trip	3

Deliverables

- ▶ Price pipeline: $\text{base} \times \text{seats} \rightarrow \text{student } 50\% \rightarrow \text{promo } \% \rightarrow 50\% \text{ prepayment}$.
- ▶ Reliability score: $\text{rating} \times \text{completion} \times (1 - \text{cancellation})$.
- ▶ Bug caught at sprint review: promo applied before student discount — fixed.

SPRINT 4

Community & Finalization

MISSION

Chat, ratings, complaints, social feed, helpdesk, multilingual interface, admin panel.

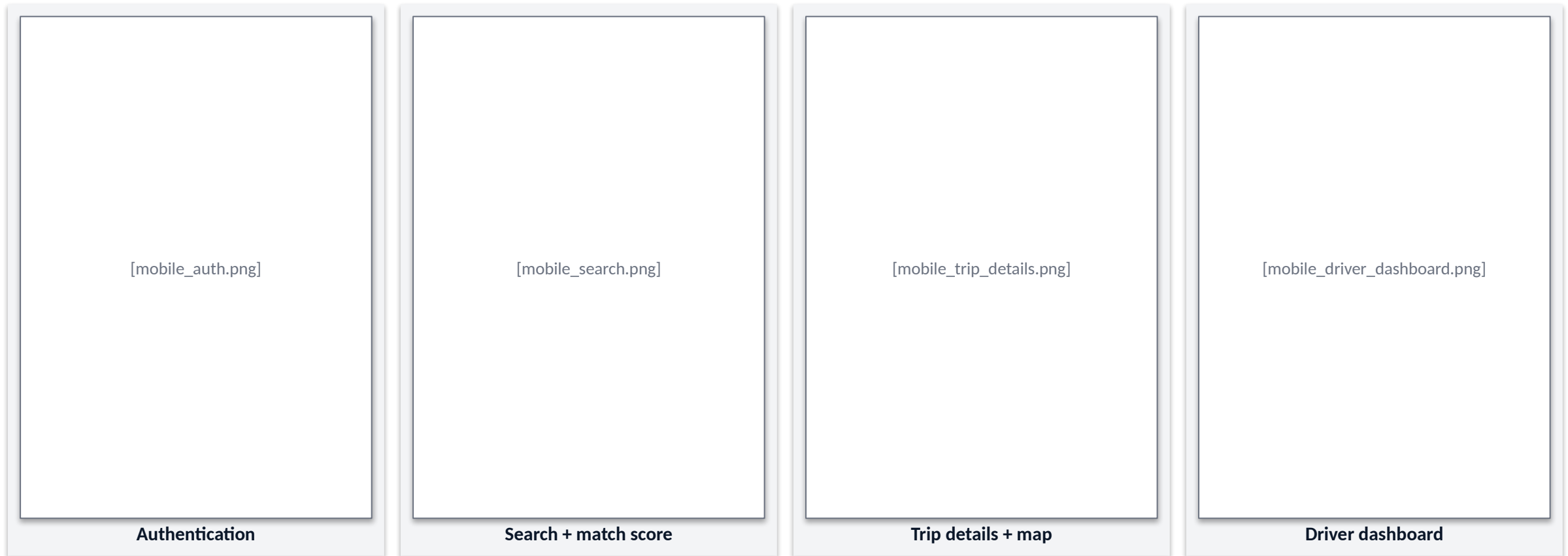
Sprint backlog (extract)

#	User story	Pts
US4.1	Real-time in-app chat (poll-based)	5
US4.2	Push notifications (FCM)	3
US4.3	Social feed (posts, likes, comments)	8
US4.4	Rate the other party after a trip	3
US4.5	File a complaint after a trip	5
US4.6	AI-assisted HelpDesk with bot escalation	5
US4.7	Switch interface FR / EN / AR (RTL)	5
US4.8	Admin panel — 7 tabs	8

Deliverables

- ▶ Chat polled every 3 s — sufficient for current scale, no WebSocket dep.
- ▶ HelpDesk bot covers 14 FAQ categories, escalates to human agent.
- ▶ Arabic flips the layout automatically through Flutter localizations.

Selected screens



Web counterparts cover the same workflows for desktop users and the administration panel.

What only the mobile makes possible

Push notifications

Firebase Cloud Messaging delivers booking, message, and ride alerts even when the app is in the background.

Interactive map

OpenStreetMap tiles + OSRM routing draw the driving line between origin and destination on each ride detail screen.

Real-time chat

Conversations linked to bookings — bubble UI, date dividers, read receipts. 3-second polling, no WebSocket dependency.

Multilingual + RTL

FR / EN / AR with automatic right-to-left layout flipping for Arabic, no manual override in any widget.

Technology stack

Web frontend

PHP 8 (server-rendered)

Routing, sessions, page rendering.

HTML5 + Bootstrap 5

Responsive layout, consistent components.

Vanilla JavaScript

Form validation, AJAX, admin tab switching.

Backend / data

PHP REST API

Front-controller under /api/, JSON responses.

PDO + bearer tokens

Parameterized queries, 32-byte hex tokens.

SQLite (WAL mode)

Shared file between web and mobile.

Mobile

Flutter (Dart)

Single codebase, native rendering on Android & iOS.

Provider + go_router

State management + declarative navigation.

FCM, flutter_map, l10n

Push, OSM map, FR/EN/AR with RTL.

Testing — continuous, not deferred

Strategy

- ▶ Unit tests on the price pipeline, prepayment, booking state machine.
- ▶ Postman collections replay end-to-end scenarios after each backend change.
- ▶ Manual cross-device tests on three Android phones.
- ▶ Weekly demo with the supervising engineer caught usability issues.

Results

Scenario	Result
Registration + student OTP	✓ Passed
Token-based session	✓ Passed
Driver application + approval	✓ Passed
Publish / cancel a ride	✓ Passed
Book with 50% prepayment	✓ Passed
Student discount + promo code	✓ Passed
Wallet top-up + history	✓ Passed
Ratings + complaints	✓ Passed
Real-time chat + push (FCM)	✓ Passed
Multilingual + RTL	✓ Passed

LIVE DEMO

ForsaDrive in action

Walk-through: registration → student verification → search → book → driver acceptance → cash collection.

Limitations & roadmap

What's missing today

- ▶ Wallet uses manual top-ups — no real payment gateway integrated yet.
- ▶ SQLite is suited for development, not high-traffic production.
- ▶ Mobile tested only on Android — iOS build not verified.
- ▶ No forgot-password email flow yet.
- ▶ Performance under concurrent load was not measured.

Next steps

- ▶ Migrate to PostgreSQL or MySQL for production.
- ▶ Integrate a Tunisian payment gateway (Konnect or Paymee).
- ▶ Deploy backend behind HTTPS on a VPS, automated backups.
- ▶ Publish the Flutter app on the Google Play Store.
- ▶ Add 2FA for driver and admin accounts + offline cache on mobile.

QUESTIONS & DISCUSSION

Thank you for your attention.

We are now ready for your questions.

Youssef BEN ABID & Anas YOUNES • ForsaDrive • 2025 — 2026